*Come play my game*

"Breathe" — The Prodigy (1994)

University of Austin

# Linear Algebra: Mathematics and Computation

---

Lecture 1 · A First Look: Something Strange Is Happening

Spring 2026

*You don't need to understand everything today. That's the point. Run the code, look at the pictures, notice what feels weird. By the last lecture, every surprise here will feel obvious.*

## §1 A Game That Started It All

In 1970, the mathematician John Conway invented a game played on an infinite grid of square cells. Each cell is either **alive** or **dead**. Every tick, each cell looks at its eight neighbours and decides whether to live or die according to a few simple counting rules.

That's it. No physics, no equations, no randomness. Just a grid, and a rule.

From this absurdly simple setup, something remarkable emerges: a tiny five-cell pattern called the **glider**. The glider moves — diagonally, one step every four ticks — forever, across the infinite grid.

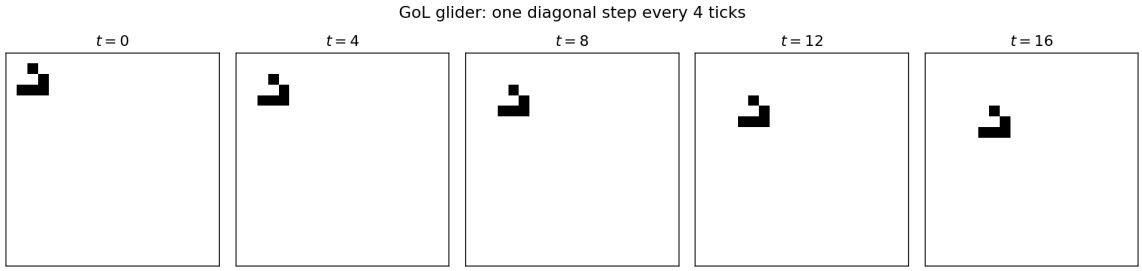


Figure 1: The GoL glider at five snapshots, each four ticks apart. The pattern shifts one cell diagonally each time.

Here is the question that will drive this entire course:

*Why diagonally? Not left, not right, not straight up.* ***Diagonally.***
*There must be something about this grid that makes the diagonal direction special.*

We will answer this question completely. But not today.

## §2 A Simpler Version

Conway's Game of Life is *nonlinear*: the rules involve counting and comparing to thresholds. Nonlinear systems are hard. Let's replace GoL with something we can actually analyse.

Instead of alive/dead, let each cell $(m, n)$ carry a real number $u_{m,n}(t)$ — think of it as an *activity level*. And instead of the threshold rule, we use a plain *linear* rule:

$$\boxed{u_{m,n}(t+1) \;=\; 1.2 \cdot u_{m+1,\,n+1}(t) \;+\; 0.8 \cdot u_{m-1,\,n-1}(t)} \tag{1}$$

In plain English: *"Next activity = 1.2 × north-east neighbour + 0.8 × south-west neighbour."*

Notice something: this rule only ever talks to the neighbours in the NE and SW directions — straight along the *main diagonal*. It completely ignores the NW and SE directions. We just built directionality into the rule by hand.

What happens when you run it? Let's find out.

## §3 Let's Experiment

### Experiment 1 — drop a single blip

Start with a 64×64 grid, all zeros except one cell in the centre set to 1. Run the rule for 20 steps.

```
import numpy as np
import matplotlib.pyplot as plt

N = 64
p, q = 1.2, 0.8

def step(u):
    # u[m,n] gets p times its NE neighbour and q times its SW neighbour
    return p * np.roll(np.roll(u, -1, axis=0), -1, axis=1) \
        + q * np.roll(np.roll(u, +1, axis=0), +1, axis=1)

# Start with a single active cell at the centre
u = np.zeros((N, N))
u[N//2, N//2] = 1.0

frames = [u.copy()]
for _ in range(20):
    u = step(u)
    frames.append(u.copy())
```

Notice anything? The signal spreads only along **one diagonal** — the north-east / south-west direction. The other diagonal carries nothing. Why?

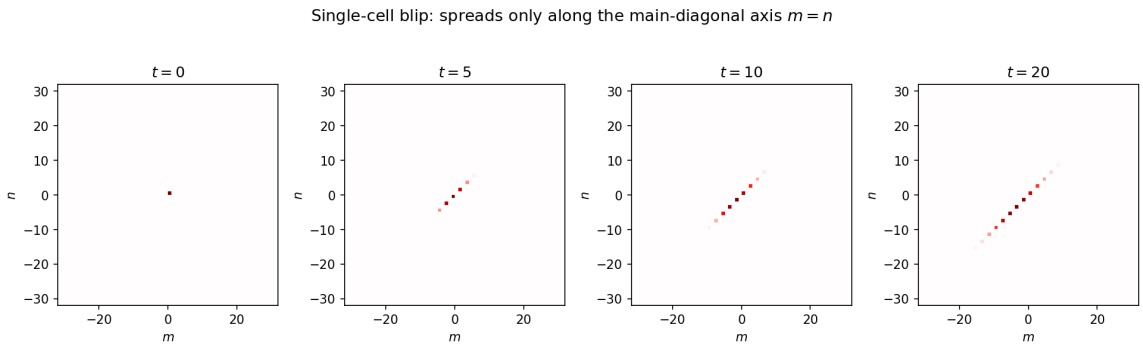


Figure 2: A single blip at the centre, evolved for 20 steps. Each frame is 4 ticks later.

## Experiment 2 — how fast does it grow?

```
energies = [np.sum(np.abs(f)) for f in frames]
for t, e in enumerate(energies):
    print(f"t={t:2d} total activity = {e:.4f}")
```

Each step the total activity **doubles**. Always. Every single step, right from the start. Yet neither 1.2 nor 0.8 equals 2. Where does 2 come from?

(Spoiler: $1.2 + 0.8 = 2$. But *why* is it the sum? That's a linear algebra story.)

## Experiment 3 — does direction matter?

Instead of a single blip, seed the grid with a wave running along each diagonal — first the *anti-diagonal* (NW–SE), then the *main diagonal* (NE–SW) — and watch what happens.

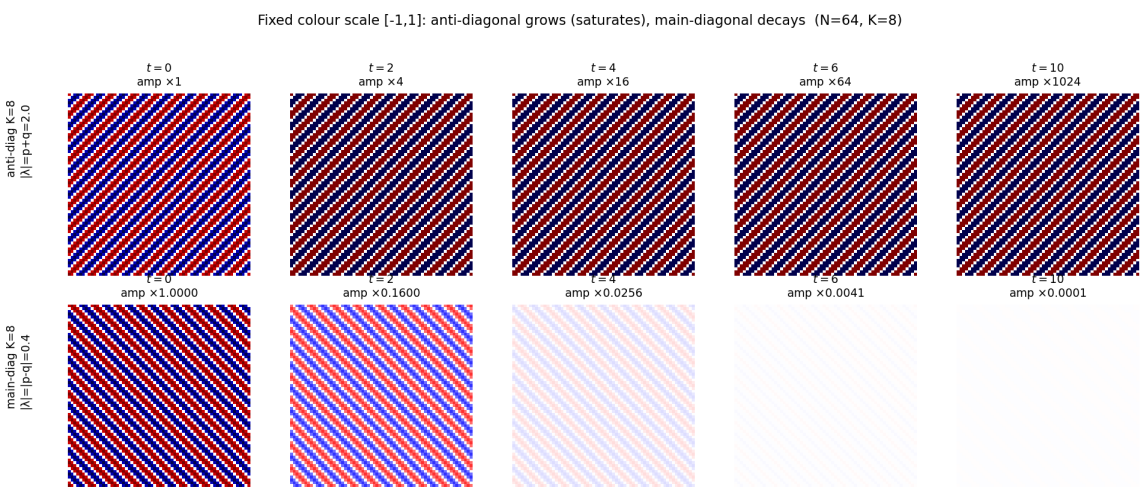


Figure 3: Top row: wave along the anti-diagonal. Bottom row: wave along the main diagonal. The anti-diagonal wave stays a perfect stripe forever (just 1024× bigger after 10 steps). The main-diagonal wave at this wavelength fades to nothing in 10 steps.

So direction does matter. Waves along the anti-diagonal all grow at exactly rate 2, regardless of wavelength. Waves along the main diagonal grow or decay depending on their wavelength. This is not a coincidence.

## §4 The Full Picture

Here is a map of every possible wave direction on the grid, coloured by how fast it grows or shrinks under one step of the rule.

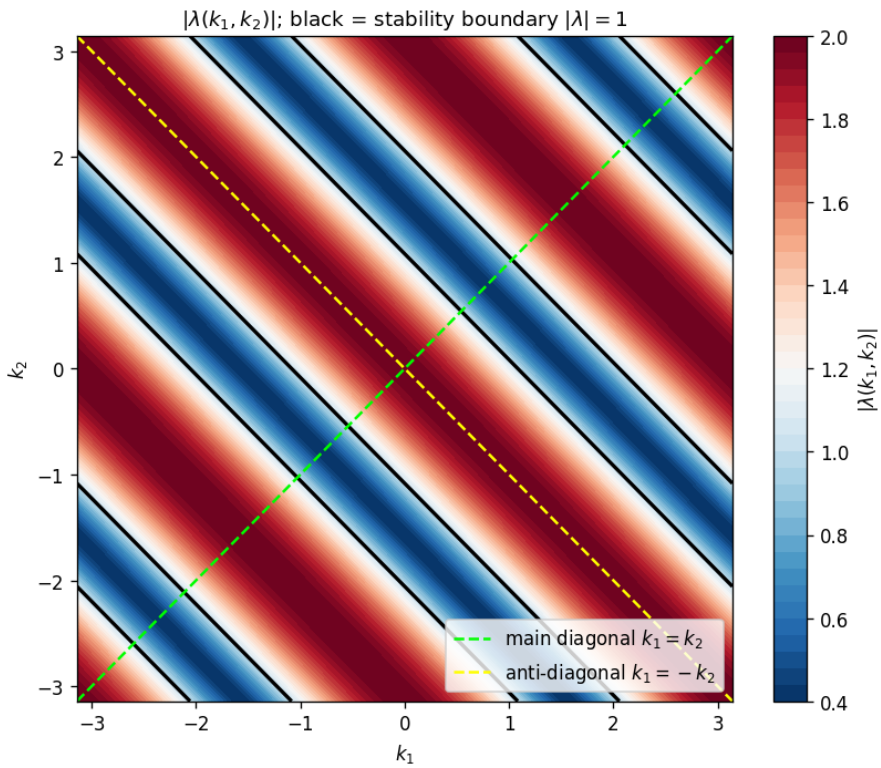


Figure 4: Growth rate $|\lambda(k_1, k_2)|$ for every possible wave direction in the grid. Black curve = boundary between growing and decaying. The anti-diagonal axis (yellow dashes) sits entirely in the 'growing' region, always at exactly 2. The main-diagonal axis (green dashes) crosses the boundary.

Each point in this picture is a wave direction. Red = growing, blue = decaying.

By the end of this course you will be able to derive this picture from scratch using two ideas: **eigenvectors** and the **Fourier transform**. For now, just look at it.

## §5 What This Course Will Teach You

Here is what the next few months are about, in plain terms.

1. **Vectors and linear maps.** The rule $u \mapsto Tu$ is a *linear map*. Linear maps are the centrepiece of the course. We will spend a long time understanding what they can and cannot do.

2. **Eigenvalues and eigenvectors.** The waves in §3 are *eigenvectors* of $T$. Their growth rates are *eigenvalues*. Once you know them, you can predict everything.

3. **Change of basis.** Switching to the 'wave basis' (the Fourier basis) turns $T$ into a simple diagonal map — multiply each component by its eigenvalue. This is why the FFT is fast.

4. **Stability.** A system is stable when all eigenvalues have magnitude $\leq 1$. The black curve in §4 is exactly the set of unstable modes.

> *The glider in Conway's Game of Life moves diagonally because the diagonal direction is the one where the linearised rule has all eigenvalues equal. Coherent structures live where all modes behave the same way.*
>
> *You are not expected to understand this sentence yet.*
> ***You will, by the end of the course.***